Customer Shopping Behaviour Analysis Report

Executive Summary

This comprehensive analysis of customer shopping behaviour leverages transactional data from 3,900 purchases across diverse product categories. By examining key demographics, purchase patterns, product preferences, and subscription behaviours, we uncover actionable insights to drive revenue growth, enhance customer loyalty, and optimize marketing strategies. Key findings include higher spending among subscribers, the impact of discounts on high-value purchases, and top-performing products by category and rating. Based on these insights, we recommend targeted subscription promotions, loyalty programs, and refined discount policies to maximize profitability. This report demonstrates my expertise in data analysis using Python, SQL, and visualization tools, making me an ideal partner for your next business intelligence project. Let's collaborate to transform your data into strategic advantages—contact me for customized solutions.

1. Project Overview

This project analyses customer shopping behaviour using a dataset of 3,900 transactional records spanning various product categories. The primary objectives are to identify spending patterns, segment customers, reveal product preferences, and evaluate subscription impacts. These insights empower businesses to make data-driven decisions, such as refining marketing campaigns, optimizing inventory, and boosting customer retention. Leveraging Python for data preparation, SQL for in-depth querying, and visualization tools for dashboards, this analysis provides a holistic view of consumer dynamics.

2. Dataset Summary

- Records: 3,900 rows
- Features: 18 columns

• Key Variables:

  - Customer Demographics: Age, Gender, Location, Subscription Status

- Purchase Details: Item Purchased, Category, Purchase Amount, Season, Size, Colour

- Shopping Behaviour: Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type

- Data Quality: No missing values were present, ensuring robust analysis without imputation needs.

3. Methodology: Exploratory Data Analysis (EDA) Using Python

Data preparation and initial exploration were conducted in Python to ensure accuracy and efficiency:

- Data Loading: Imported the dataset using Pandas for seamless handling.

- Initial Exploration: Utilized `df.info()` to assess structure and `df.describe()` for summary statistics, identifying distributions and potential outliers.

count	3900.000000	3900.000000	3900.000000	3900.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.749949	25.351538
std	1125.977353	15.207589	23.685392	0.716223	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.700000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

- Column Standardization: Renamed columns to snake_case for improved readability and consistency in code.

- Feature Engineering:

- Binned customer ages into meaningful groups (e.g., 18-24, 25-34) to facilitate segmentation.

- Derived a `purchase_frequency_days` column to quantify buying intervals from transactional dates.

- Data Consistency Checks: Analyzed redundancy between `discount_applied` and `promo_code_used`; removed the latter to streamline the dataset.

- Database Integration: Connected the Python script to PostgreSQL and loaded the cleaned DataFrame for advanced SQL-based querying.

This phase ensured the data was primed for insightful analysis, highlighting my proficiency in Python-based ETL (Extract, Transform, Load) processes.

4. In-Depth Data Analysis Using SQL

1. Leveraging PostgreSQL, we addressed critical business questions through targeted queries. Below are the key analyses and findings:

Revenue from male customers is 2.1 times higher than that from female customers (67.7% vs. 32.3% of total revenue), revealing a significant growth opportunity through targeted strategies aimed at increasing female participation.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. High-Spending Discount Users: Identified 450 customers who applied discounts yet spent above the average purchase amount ($59.50), representing a segment ripe for upselling.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90

3. The top 5 products have very close ratings (3.78–3.86), with Gloves (3.86) and Sandals (3.84) slightly ahead, indicating solid overall satisfaction. T-shirts (3.78) suggest an opportunity for product or merchandising improvement.

	item_purchased text	Avarage Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.81
4	Hat	3.81
5	T-shirt	3.78

4. Customers with Express shipping spend 3.5% more ($60.48 vs. $58.46 for Standard). This shows a higher-value group per order and a great opportunity to boost revenue by encouraging faster shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. Subscribers make up 27% of customers and contribute ~27% of total revenue, with nearly identical average spend ($59.49 vs $59.87 for non-subscribers). The biggest growth lever is increasing subscription adoption.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. The products most reliant on discounts are Hat (50%), Sneakers and Coat (49%), Sweater (48%), and Pants (47%). High discount rates suggest strong price sensitivity and an opportunity to refine promotional strategies to protect margins.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

7. Nearly 80% of customers are Loyal, while only 2.1% are New. Retention is excellent, but acquiring new customers is the key area of focus and the main opportunity for growth.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. Clothing and Accessories lead sales (Blouse, Pants, and Jewelry each with 171 orders). The top 10 items account for ~41% of orders — prioritizing stock, promotions, and cross-selling in these products is the key growth driver.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163

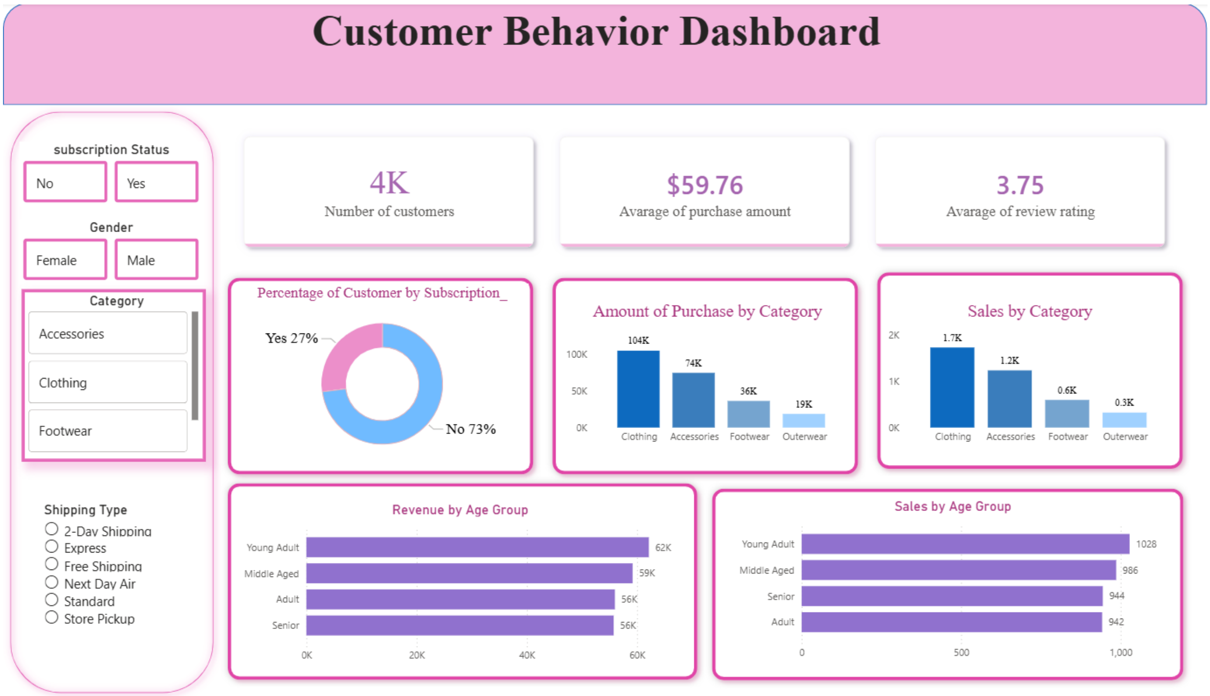
9. Subscribers show a slightly higher repeat buyer rate (91.0% vs. 88.4% for non-subscribers). While the subscription provides a modest boost to retention, overall purchase experience and product quality appear to be the dominant drivers of repeat purchases.

	subscription_status text	repeat_buyears bigint
1	No	2518
2	Yes	958

10. Revenue is evenly distributed across age groups, with Young Adults slightly leading at 26.7% ($62,143), followed by Middle Aged (25.4%). Multi-generational strategies are key to maximizing growth.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle Aged	59197
3	Adult	55978
4	Senior	55763

5. **Interactive Dashboard: Customer Behaviour and Shipping Insights**



6. **Business Recommendations**

Based on the analysis, the following strategies are recommended to enhance business performance:

- **Boost subscriptions** — Only 27% of customers are subscribers; offer free express shipping and exclusive discounts to convert non-subscribers and increase recurring revenue.
- **Strengthen new customer acquisition** — With just 2% of the base being new, invest in social media ads, micro-influencers, and aggressive first-purchase offers.
- **Review discount strategy** — Items like Hat (50%) and Sneakers/Coat (49%) rely heavily on high discounts; gradually reduce them and prioritize bundles and added value to protect margins.

- **Focus on Clothing and Accessories** — Dominant categories; prioritize stock, cross-selling, and campaigns for top items (Blouse, Pants, Jewelry, Sunglasses).
- **Leverage balanced age distribution** — Young Adult leads slightly (27%); maintain multi-generational strategies with a light emphasis on youth and comfort for seniors.
- **Improve reviews (average 3.75)** — Encourage post-purchase reviews and enhance photos/descriptions to boost satisfaction and conversion.

Immediate priorities: subscriptions + new customer acquisition + discount review.

7. Conclusion

This analysis not only illuminates key drivers of customer behaviour but also provides a roadmap for sustainable growth. By integrating Python, SQL, and visualization expertise, I've delivered insights that can transform your business operations. As a freelance data analyst with a proven track record in e-commerce analytics, I'm eager to apply these skills to your projects.

For interactive dashboard access or custom analyses, please contact me. Denilsonfranciscocuamba@gmail.com